

expectation) for all covariates used to estimate the propensity score and in all bins where there are both visited and not visited doctors.

The process of assessing balance was conducted for all variables and all bins and considered adequate in the sense that it was considered plausible

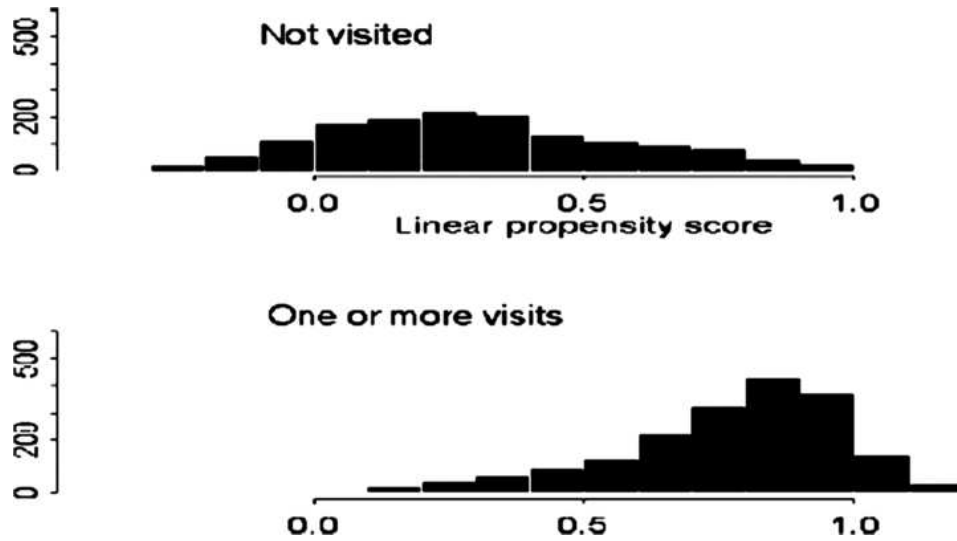


FIG. 3. Histograms for summarized background variables: linear propensity score.

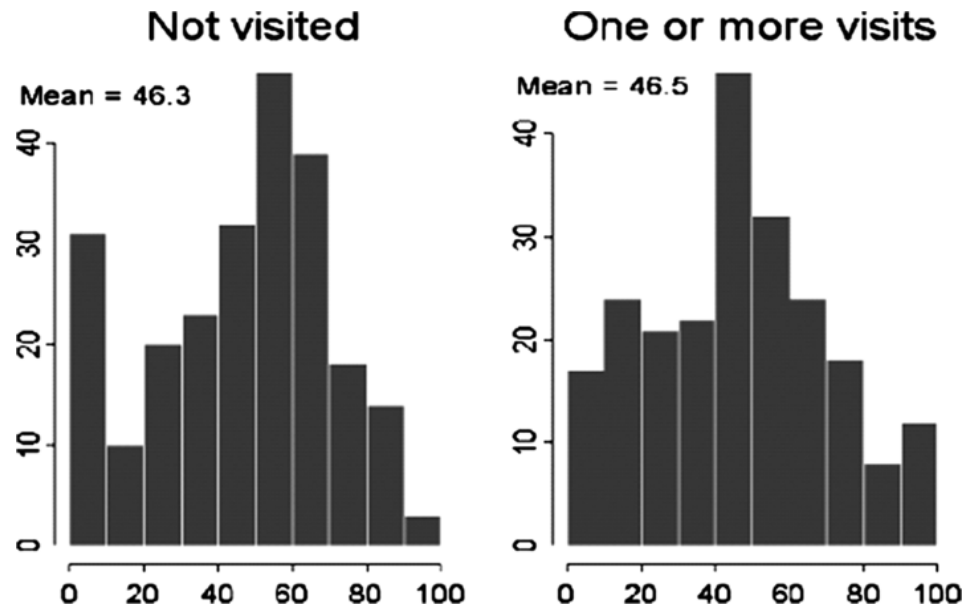


FIG. 4. Histograms for a variable in a subclass of propensity scores: prior Rx score.

that a randomized block experiment had become reconstructed, except for the bins with only visited or not visited doctors. Admittedly, there is an aspect of “art” operating here, in that random imbalance of prognostically important covariates (i.e., ones thought to be strongly related to outcome variables) was considered more important to correct than more extreme imbalance in prognostically unimportant ones, but the field of statistics will always benefit from scientifically informed thought. Nevertheless, better guidance on how to conduct this process more systematically is needed, and is in development; see, for example, Imbens and Rubin [(2008b), Chapters 13 and 14].

In any case the design phase was complete, except for the specification of model-based adjustments to be made within the bins, and the more detailed analyses used to rank doctors by priority to visit. Readers interested in the conclusions, which are a bit surprising, should check [Rubin and Waterman \(2006\)](#).

5. A principal stratification example.

5.1. *The causal effect of being treated in large volume versus small volume hospitals.* The third example illustrates the point that the design phase in some observational studies may involve conceptualizing the hypothetical underlying randomized experiment that lead to the observed data as being more complex than a randomized block or randomized paired comparison. In particular, in some situations, we may have to view the hypothetical experiment as being a randomized block with noncompliance to the assigned treatment, a so-called “encouragement” design [Holland (1988)]. In many

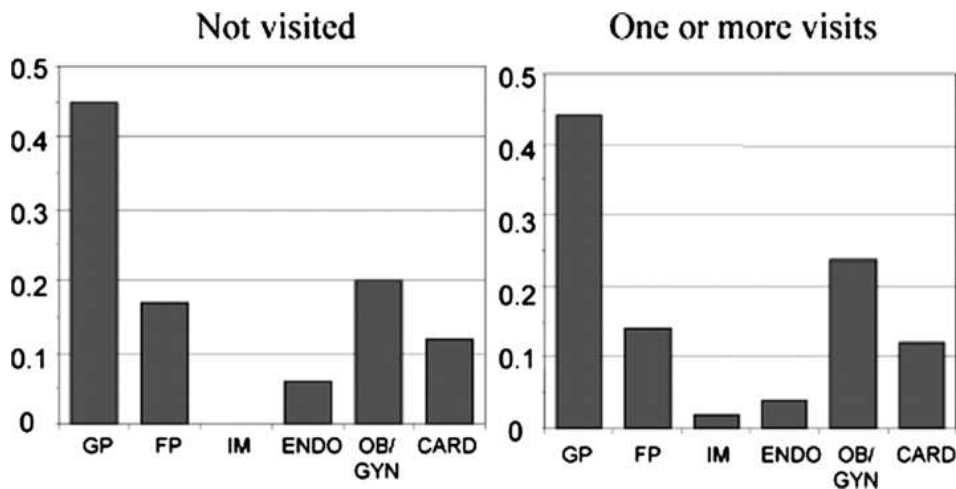


FIG. 5. Histograms for a variable in a subclass of propensity scores: specialty.

settings with human subjects, even an essentially perfectly designed randomized experiment only randomly assigns the encouragement to take treatment or control because we cannot force people to take one or the other. In the context of a perfectly double-blind experiment, where the subjects have no idea whether they are getting treatment or control, there will be no difference in compliance rates between the treatment versus control groups, but there are often side effects that create different levels of compliance in the conditions [Jin and Rubin (2008)]. In such cases, the ideas behind “instrumental variables” methods [Angrist, Imbens and Rubin (1996)] as generalized to “principal stratification” [Frangakis and Rubin (2002)] can be very useful.

5.2. *Propensity score subclassification for diagnosing hospital type.* We illustrate this design using a small observational data set from the Karolinska Institute in Stockholm, Sweden. Interest focuses on the treatment of cardia cancer patients in Central and Northern Sweden, and whether it is better for these patients to be treated in a large or small volume hospital, where volume is defined by the number of patients with that type of cancer treated in recent years. The data set has 158 cardia cancer patients diagnosed between 1988 and 1995, 79 diagnosed at large volume hospitals, defined as treating more than ten patients with cardia cancer during that period, and 79 diagnosed at the remaining small volume hospitals. These sample sizes are small, but the data set is the only one currently available in Sweden to study this important question.

Generally, the commonly held view is that being treated in a large volume hospital is better, but the opposite argument could be made when the large volume treating hospital is far from a support system of family and friends, which presumably may be more available in small volume hospitals. The most critical policy issue concerns whether the cardia cancer treatment centers at small volume hospitals can be closed without having a deleterious effect on patient survival. If so, resources could be saved because patients diagnosed at small volume hospitals could be transferred to large volume treating hospitals, and if it is true that large volume cardia cancer treatment centers offer better survival outcomes, then the small volume ones should arguably be closed in any case. Our data set has hospital volume and patient survival information in it.

Because of the uniform training of doctors under the socialized medical system in Sweden, the assignment of large versus small “home hospital type,” where the cancer was diagnosed, was considered by medical experts to be unconfounded, that is, essentially assigned at random within levels of measured covariates, X : age at diagnosis, date of diagnosis, sex of patient and urbanization. The decision maker is the individual patient, so our

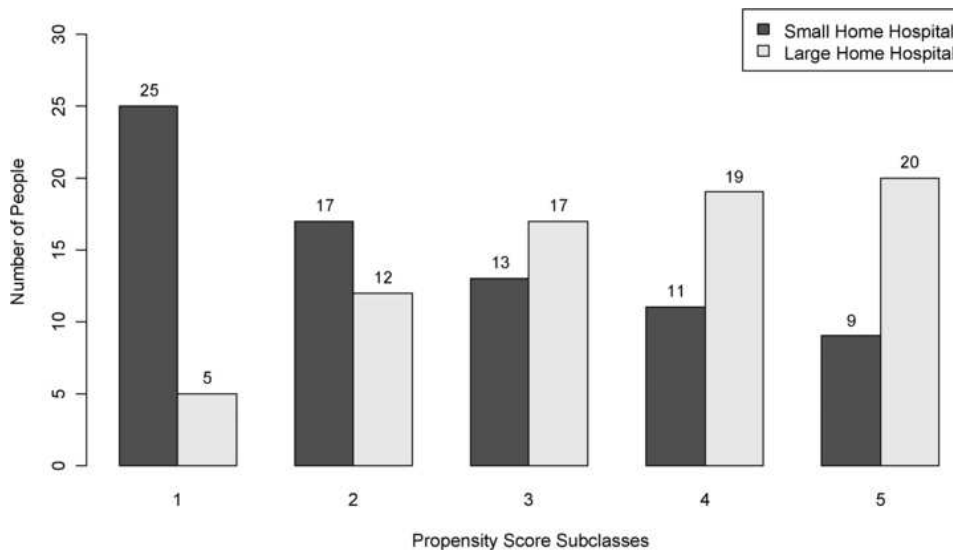


FIG. 6. *Cardia cancer, number of people, subclassified by propensity score.*

dataset seems well-suited for studying the causal effect of home hospital type on survival.

Propensity score analyses were done to predict diagnosing (home) hospital type from X , including nonlinear terms in X . It was decided that the age of patient should be limited to between 35 and 84 because the two patients under 35 (actually both under 30) were both diagnosed in large volume hospitals, and longer term survival in the 8 cardia cancer patients 85 and over was considered unlikely no matter where treated, and would therefore simply add noise to the survival data. Propensity score analyses on the remaining 148 patients led to five subclasses; these are summarized in Figures 6–8 are “Love plots” [Ahmed et al. (2006)] summarizing balance before and after this subclassification, for binary and continuous covariates, respectively.

5.3. *Treating hospital type versus home hospital type.* If patients were always treated in the same hospital where they were diagnosed, estimating the causal effects of hospital type would now be easy because of the assumed unconfounded assignment of diagnosing hospital type. However, there are transfers between hospital types, typically from small to large—33 of the 75 diagnosed in a small hospital transferred to a large one for treatment, but sometimes from large to small—2 of 75 transferred this direction. The reasons for these transfers are considered quite complex. The decisions are made by the individual patient, but clearly with input from doctors, relatives, and friends, where the issues being discussed include speculation about